

Skill Nexus Week 4 – End-to-End Capstone Report

Project: EDA, Sales Analysis, Regression Modeling & Dashboard Preparation

Metric	Result
Original rows	700
Rows after cleaning	700
Columns	16
Missing Discount Band values handled	53
Duplicate rows removed	0
Total Sales	\$118,726,350.26
Total Profit	\$16,893,702.26
Regression test R ²	0.7828

Data Cleaning

Column names and text fields were standardized, missing Discount Band values were labeled 'Not Specified', the Date field was converted to a proper date type, and exact duplicate rows were checked and removed when present. Two derived fields were added: Profit Margin % and Discount %.

Regression Model

A linear regression pipeline was trained with an 80/20 train-test split (random_state=42). The model uses units sold, sale price, discounts, manufacturing price, time fields, and categorical business fields. The test-set R² was 0.7828. The model should be treated as a planning aid and validated further before operational use.

Stakeholder Recommendations

- Prioritize the strongest sales segments and countries in sales planning.
- Use high-performing products as anchors for campaigns and cross-selling.
- Monitor profit margin together with sales to avoid growth driven by excessive discounting.
- Review records with large prediction errors before relying on the model for decisions.
- Use the prepared KPI, monthly, segment, country and product tables to build the required Power BI dashboard.